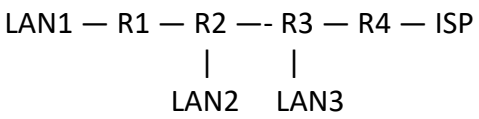
Lab 2: Supernetting in a Linear Chain with Unequal LAN Sizes

Objective:

Build a 4-router chain where the last 3 routers each contain LANs that must be aggregated upstream.

Constraints:

1. R3 LAN size = (R1 LAN size)
2. R2 LAN size = 2 × (R1 LAN size)
3. LAN1, LAN2, LAN3 must form a **naturally-aligned continuous block**.
4. R4 must advertise **only one supernet** toward ISP.



Physical links: R1–R2, R2–R3, R3–R4 (each a point-to-point link).

Tasks:

- Assign LAN1, LAN2, LAN3.
- Determine aggregate mask.
- Configure R4 to summarize toward ISP.
- Enable static routing or RIP.
- Verify that R1 only receives the supernet route.

Use: Cisco 2811 Router

Because:

- It supports **WIC-2T** or **WIC-1T** serial interface modules
- It has **2 built-in Fast Ethernet ports (Fa0/0 and Fa0/1)**
- After adding **one WIC-2T card**, you get **2 serial ports (S0/0/0 & S0/0/1)**
- Perfect for "square-shape" router-to-router serial connections